

RFB–HORN: Learnable oscillatory frequency decomposition for compact spoken-digit recognition

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github.com/samim23/oscnet · July 2026

Abstract

Speech carries structure across timescales. Inspired by AudioPrism [2], we study a small digital stack with two stages: a learnable *resonator filter bank* (RFB) and a densely coupled *HORN* (harmonic oscillator recurrent network). Biology is inspiration only.

On Google Speech Commands digits (v0.02), learnable RFB→dense HORN reaches 0.76 ± 0.02 clean accuracy at $\sim 7k$ parameters, while the same RFB features with a pooled MLP collapse to 0.36 ± 0.01 —so *temporal integration* is load-bearing. Matched sequential controls (including a GRU) show that oscillation is not the only way to use time—but on mel features, dense HORN nearly matches a $\sim 3\times$ larger GRU ($\sim 7k$ vs. $\sim 19k$ params), and RFB arms retain an additive-noise advantage under clean training. We also report structure ablations (equal-frequency $\approx$ chance), learning requirements, robustness tradeoffs, and nulls (phase, AGC, high→low coupling under our probe).

1 Introduction

Speech encodes information across timescales. Biological hearing is often described as combining tonotopic decomposition with multi-scale integration [2]. We treat that description as *inspiration* for a small digital encoder—not as neuroscience validation.

AudioPrism [2] established that a *frozen* cochlear-like resonator bank (PRISM) in front of a HORN unlocks learnability for small spoken-digit models. We extend that two-stage idea (Figure 1) with learnable $\{\omega, Q\}$, framed features, and dense learned coupling, and we include matched sequential controls so gains are not confused with “any recurrence helps.”

Contributions.

1. **Temporal integration is load-bearing.** Pooled MLP on RFB frames fails; framed HORN succeeds.
2. **HORN is compact.** At $\sim 7k$ parameters, mel→dense HORN nearly matches a larger non-oscillatory sequential head on clean digits (Section 4.2).
3. **RFB structure and learning matter on the RFB path.** Equal-frequency banks fail; frozen RFB trails mel; learning $\{\omega, Q\}$ lifts RFB→HORN (Figure 3).
4. **Noise asymmetry.** Clean-trained RFB sequential arms retain higher white-noise accuracy than matched mel arms in our protocol.
5. **Nulls.** Phase readout, bank AGC, and high→low coupling streamlining fail under our probes.

RFB-HORN for compact spoken-digit recognition

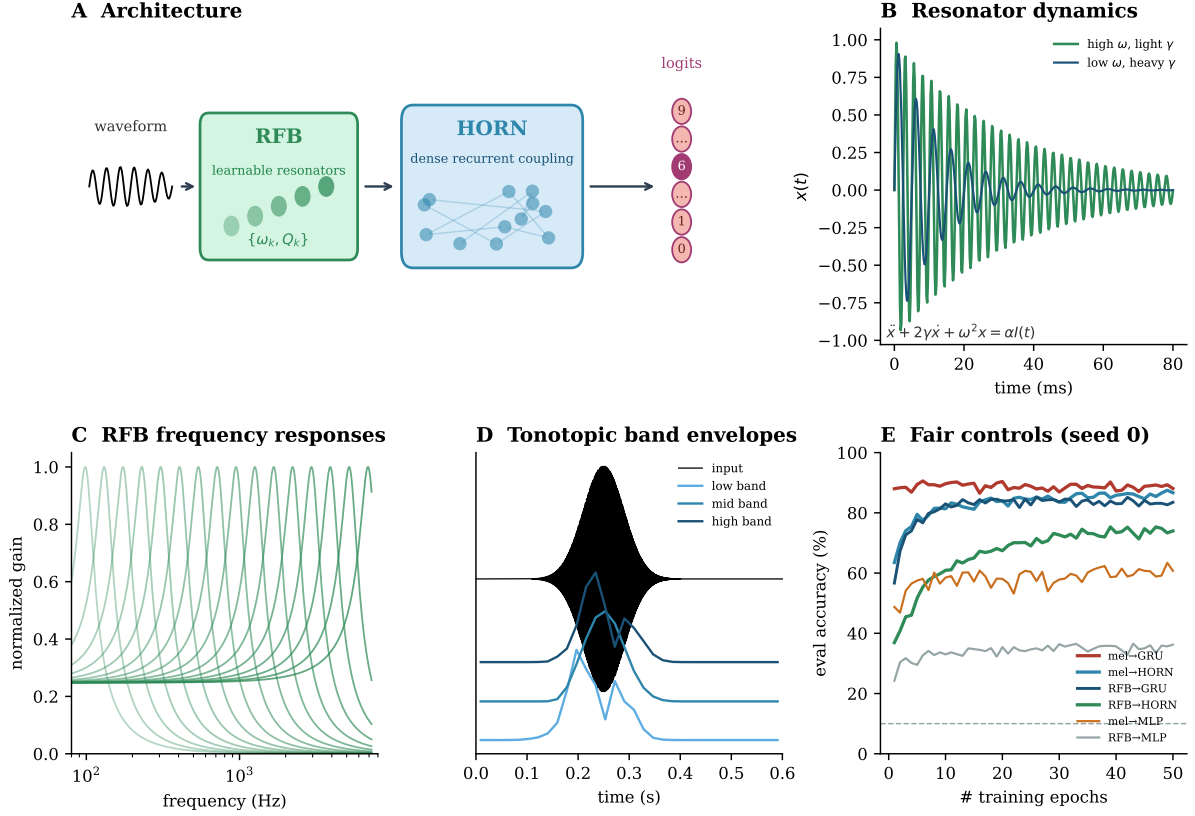


Figure 1: **(A)** Two-stage stack: learnable RFB into densely coupled HORN. **(B)** DHO impulse responses. **(C)** Log-spaced RFB gains. **(D)** Tonotopic band envelopes. **(E)** Fair-control training curves (seed 0): sequential heads dominate pooled MLP; mel leads on clean accuracy.

2 Related work

AudioPrism. Pietras et al. [2] stack a frozen analytic resonator bank with a trainable HORN. We cite that as prior art—not as an identical protocol we claim to beat. Our extensions: learnable $\{\omega, Q\}$, framed features, dense W , classical and sequential controls, robustness, and nulls.

Learnable filterbanks. SincNet [3] and LEAF [6] show constrained learnable fronts help in small-channel / noisy regimes. We use a driven DHO power response—physics-*shaped*, not hardware.

Controls. We report exact trainable parameter counts per arm and compare pooled MLP, framewise GRU, and HORN under RFB and mel frontends. Dense W adds parameters relative to fractal-fixed HORN; the default GRU head is larger ($\sim 19k$). Section 4.3 adds an iso-parameter GRU ($H=34$, $\sim 7k$).

3 Method

3.1 Resonator filter bank

Each band k obeys

$$\ddot{x}_k + 2\gamma_k \dot{x}_k + \omega_k^2 x_k = \alpha_k I(t), \quad \gamma_k = \omega_k / (2Q_k). \quad (1)$$

Default init: $K=32$, $Q=4$, 16 kHz, log-spaced $f \in [100, 7000]$ Hz, unit-peak α_k . Learnable banks optimize bounded residuals $\Delta \log \omega_k, \Delta \log Q_k$ with $\log \omega = \log \omega_0 + 0.5 \tanh(\Delta \log \omega)$ (likewise Q), then sort and clip ($Q \in [0.5, 64]$, ω below $0.9 \times$ Nyquist). Gradients use the frequency-domain DHO power response (not BPTT through an IIR). Features are $T=16$ non-overlapping frames of log-power amplitude

$$z_{t,k} = \log(\langle x_k^2 \rangle_{\text{frame } t} + \varepsilon), \quad (2)$$

with band-spacing regularizer weight 0.1 when learning $\{\omega, Q\}$.

3.2 Heads

MLP. Two-layer GELU on *mean-pooled* features.

GRU. Standard GRU [1] (`eqx.nn.GRUCell`) over frames; linear readout on the final state.

HORN. Input $u_t = W_{\text{in}} z_t$, recurrent $r_t = W v_t$, and fixed-dynamics nonlinear oscillators ($\omega_h = 2\pi/28$, $\gamma_h = 0.01$, $\alpha = 0.04$, $\Delta t = 1$ per frame):

$$x \leftarrow x + \Delta t v, \quad (3)$$

$$v \leftarrow v + \Delta t (\alpha \tanh(u_t + r_t) - \omega_h^2 x - 2\gamma_h v), \quad (4)$$

with linear readout on x_T . Trainable: W_{in} , W , readout. Coupling W : fractal-fixed, dense, or dense-tonotopic init.

3.3 Protocol

Data. Speech Commands v0.02 [4] digits (**zero-nine**); 16 kHz, peak-normalized 1.0 s clips. Mel uses the full-rate waveform; the RFB frontend uniformly subsamples each clip to 4096 samples and scales its internal sample rate accordingly (compute tradeoff; band schedule stays $[100, 7000]$ Hz in that scaled rate). Official `validation_list.txt` and `testing_list.txt` are pooled as the evaluation pool; remaining files form the train pool (speaker disjoint by construction of those lists). We draw balanced subsamples $N_{\text{train}}=10^4$, $N_{\text{eval}}=1200$ with seed-dependent shuffles; mid-scale confirm uses 4k train. Training uses a fixed 50-epoch schedule with no early stopping; we report final-epoch evaluation accuracy (the evaluation pool was also used for iterative inspection during the project).

Opt. Adam, batch 64, 50 epochs; lr 3×10^{-3} (RFB) or 10^{-2} (mel); four seeds $\{0, 1, 2, 3\}$ unless noted.

Noise. Optional train gain ± 12 dB and, with prob. p , one of white / pink ($1/f$) / bandlimited (300–3000 Hz) noise at SNR $\mathcal{U}[0, 15]$ dB. Eval noise: independent 10 dB draws (same generators) on the evaluation subsample. Unless noted, Table 2 models are *clean-trained* ($p=0$).

4 Experiments

Chance = 0.10. Accuracies are mean $\pm$ sample std over seeds.

4.1 Pooled controls understate temporal heads

Table 1 reproduces the early headline: RFB $\rightarrow$ HORN clears pooled mel+MLP, and pooled MLP on RFB frames collapses. Dense W recovers ~ 11 points over frozen fractal structure. This comparison alone cannot attribute the gain to *oscillation* rather than *sequence use*.

4.2 Fair sequential controls (four seeds)

Table 2 and Figures 1 / 2 clarify what the architecture is doing:

- Sequential heads (HORN or GRU) dominate pooled MLP.

Table 1: Coupling / pooled-head study (two seeds). Clean is mean $\pm$ sample std; noise columns are seed means. MLP mean-pools frames.

Model	Params	Clean	White	Pink	Band
RFB $\rightarrow$ dense HORN	7083	0.761 $\pm$ 0.029	0.469	0.634	0.372
RFB $\rightarrow$ tonotopic HORN	7083	0.758 $\pm$ 0.017	0.485	0.651	0.382
RFB $\rightarrow$ fractal-fixed HORN	2987	0.649 $\pm$ 0.035	0.247	0.437	0.248
mel $\rightarrow$ MLP (pooled)	5514	0.608 $\pm$ 0.014	0.398	0.491	0.391
RFB $\rightarrow$ MLP (pooled)	5738	0.356 $\pm$ 0.004	0.175	0.333	0.153

Table 2: Fair-control package (10k train, four seeds; clean-trained).

Frontend	Head	Params	Clean	White 10 dB
mel	GRU	19338	0.880$\pm$0.019	0.381 $\pm$ 0.007
mel	dense HORN	6859	0.863 $\pm$ 0.008	0.310 $\pm$ 0.055
RFB learn	GRU	19562	0.836 $\pm$ 0.011	0.646$\pm$0.051
RFB learn	dense HORN	7083	0.764 $\pm$ 0.020	0.465 $\pm$ 0.027
mel	MLP (pooled)	2762	0.600 $\pm$ 0.007	0.378 $\pm$ 0.047
RFB learn	MLP (pooled)	5738	0.356 $\pm$ 0.009	0.231 $\pm$ 0.053

- On clean digits, a larger GRU is slightly ahead; HORN stays close at $\sim 1/3$ the parameters (mel: 0.86 vs. 0.88).
- RFB sequential arms keep more white-noise accuracy under clean training than matched mel arms.
- With fair heads, mel remains competitive with RFB on clean digits.

4.3 Iso-parameter GRU ($\sim 7k$)

Matching parameter budgets (Table 3) does not reverse the story: on mel, HORN stays within ~ 1 point of a $\sim 7k$ GRU; on RFB, the iso-parameter GRU still leads on clean accuracy and white-noise robustness. Compactness remains a HORN strength on the mel path; uniqueness does not.

4.4 Structure and learning on the RFB path

Before learning, Stage 0 (MLP heads) already shows frequency structure matters:

Frontend	Accuracy
mel / STFT / conv1d	~ 0.58 – 0.61
frozen log RFB	~ 0.44
equal-frequency RFB	~ 0.14 – 0.16
raw / raw-wide	~ 0.13

Four-seed mid-scale confirm: learnable RFB $\rightarrow$ HORN 0.648 ± 0.025 on digits and 0.608 ± 0.033 on core-10, vs. mel $\rightarrow$ MLP 0.583 ± 0.027 / 0.473 ± 0.009 (pooled-head protocol).

4.5 Clean vs. noise augmentation

For RFB $\rightarrow$ dense HORN (two seeds), soft aug ($p=0.4$, ± 12 dB) moves clean $0.761 \pm 0.029 \rightarrow 0.740 \pm 0.006$ while white 10 dB rises $0.469 \pm 0.019 \rightarrow 0.642 \pm 0.035$ (Figure 4). Harder aug ($p=0.7$) reaches white 0.678 ± 0.022 at a larger clean cost (0.705 ± 0.021).

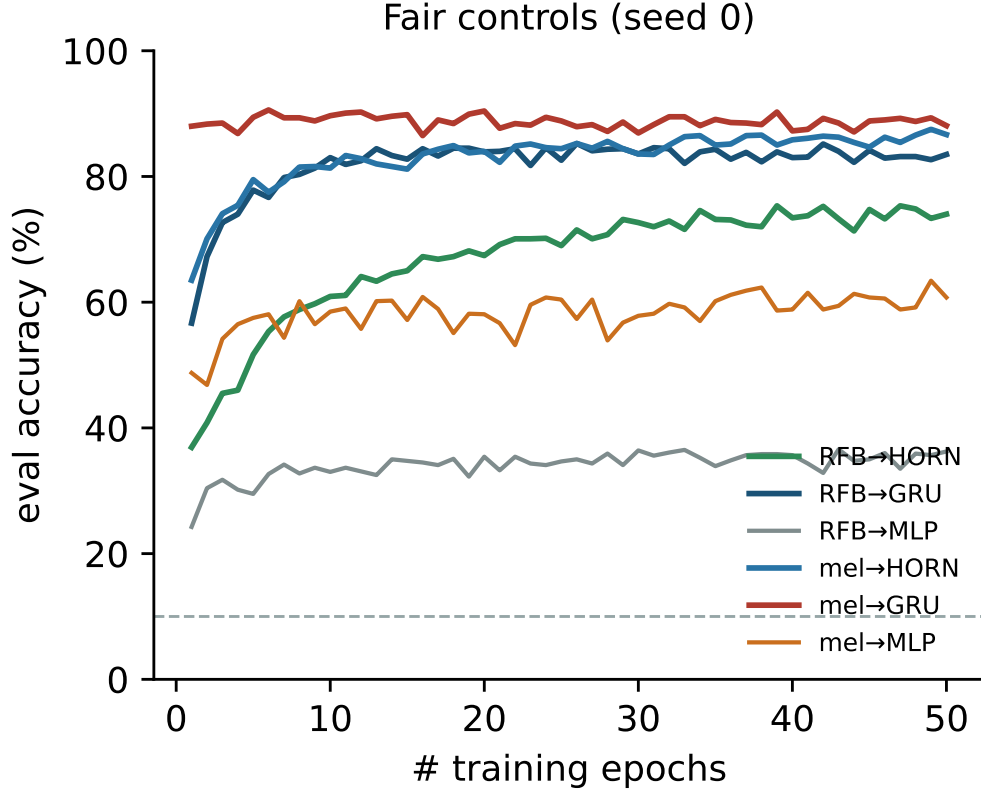


Figure 2: Fair-control training curves (seed 0). Sequential heads dominate pooled MLP; mel+GRU/HORN lead on clean accuracy.

Table 3: Iso-parameter sequential heads (10k train, four seeds; clean-trained). GRU hidden size $H=34$ vs. dense HORN $H=64$.

Frontend	Head	Params	Clean	White 10 dB
mel	GRU ($H=34$)	7218	0.873 ± 0.016	0.329 ± 0.088
mel	dense HORN	6859	0.863 ± 0.008	0.310 ± 0.055
RFB learn	GRU ($H=34$)	7442	0.845 ± 0.010	0.615 ± 0.018
RFB learn	dense HORN	7083	0.764 ± 0.020	0.465 ± 0.027

4.6 What failed

Relative to AudioPrism-style stories. Frozen RFB does not beat mel here; high→low coupling ratios stay ≈ 1.0 – 1.06 under our preferred-band probe (Appendix A). We do not claim a contradiction of AudioPrism hardware results.

Additional nulls (Figure 5): phase-only ~ 0.18 ; AGC $\rightarrow$ chance; tonotopic init $\approx$ dense on accuracy without creating high→low structure.

5 Discussion

The corrected claim is narrower—and stronger scientifically—than “RFB beats mel.” **Pooled readouts hide the value of temporal integration.** With a framed sequential head, mel remains strong on clean digits, while RFB shows a noise-robustness advantage in our clean-train stress tests. HORN’s practical niche here is **compact sequence modeling** (Table 3)—competitive accuracy at small parameter count, not unique supremacy over every recurrent

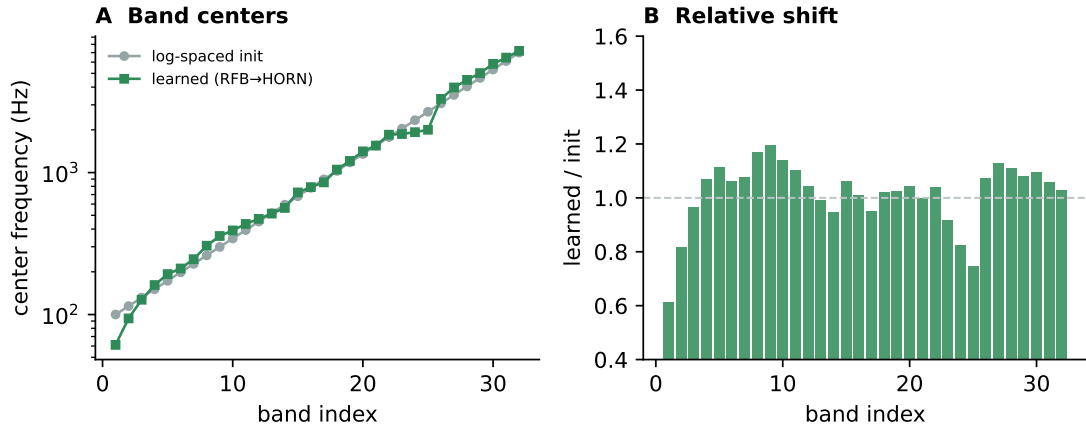


Figure 3: Illustrative learned RFB centers after RFB→dense HORN training (seed 0). Bands remain roughly log-spaced with mean $|\log(\omega/\omega_0)| \approx 0.10$.

alternative.

Limitations. Digit-only Speech Commands; pooled val+test evaluation subsample (not the full official test list); evaluation pool was inspected during development (no held-out final blind test); RFB uses 4096-sample waveform subsampling while mel does not (Section 3); no full 35-class KWS, large SSL, or tiny-CNN baselines; biological language is inspirational only.

Default recipe (when choosing HORN). Learnable RFB or mel frames, dense HORN, $K=32$, $H=64$, 16 steps; soft aug if noise robustness matters.

6 Conclusion

Learnable resonator decomposition with oscillatory recurrence is a viable small spoken-digit encoder when frames are integrated over time. HORN is a compact sequential head; RFB contributes a noise-robustness role; several biologically flavored extras fail cleanly. That is a falsifiable extension of the AudioPrism architectural idea—not a claim of ASR supremacy.

Acknowledgments

We thank the AudioPrism authors for a clear architectural target. Experiments ran on Modal A10G GPUs. AI research agents assisted with implementation, sweeps, analysis, and drafting under the author’s direction; the author is solely responsible for the scientific claims. Code and CSV summaries: github.com/samim23/oscnets [5].

References

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- [2] M. Pietras, D. Carvalho, D. Dubinin, H. Ferrand, W. Singer, and F. Effenberger. AudioPrism: A resonator–HORN architecture for spoken digit recognition. In *International Conference on Neuromorphic Computing and Engineering (ICNCE)*, 2026. Extended abstract / contribution.

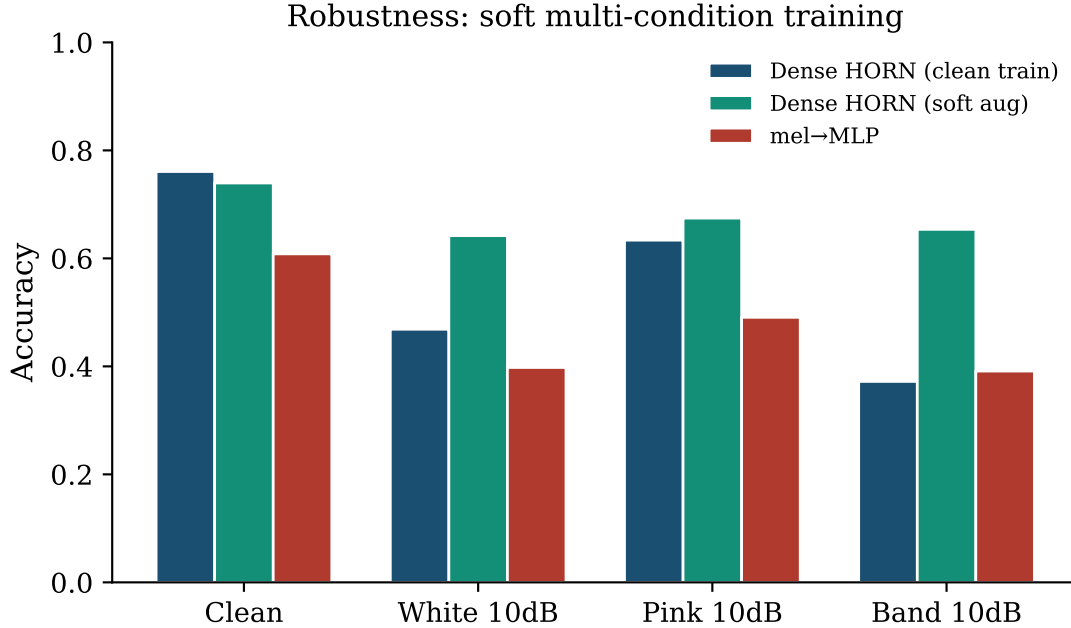


Figure 4: Soft augmentation trades a little clean accuracy for large additive-noise gains (RFB→dense HORN).

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A Coupling diagnostics

Table 4: Coupling diagnostics (scale, clean train; two seeds).

Coupling	High→low ratio	Ablate-low drop	Ablate-high drop
dense	1.04	0.63	0.55
dense tonotopic	1.01	0.63	0.56
fractal fixed	1.00	0.54	0.45

B Reproducibility

Sweep entrypoint: `scripts/modal_audio_digit.py (-sweep-preset)`. Key presets: controls, isoparam, scale, noiseaug, tonotopic (prefix `rfb_plus_`). Summaries: `outputs/analysis/`.

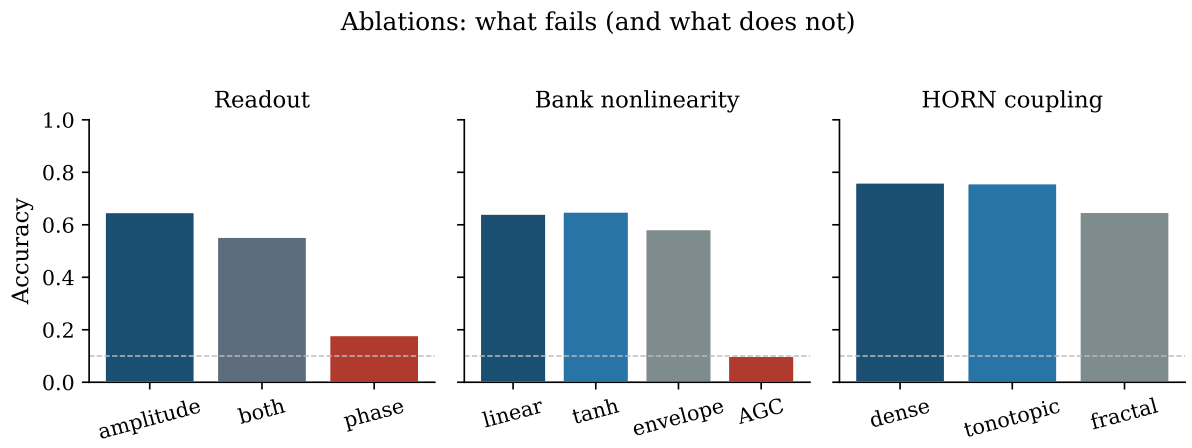


Figure 5: Ablations: amplitude wins; blunt nonlinearities fail; dense coupling beats frozen fractal structure.